**COMP5400 Coursework 2**

# Phase Transitions and Criticality in Conway's Game of Life

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## AI Usage Declaration

**Classification: Amber**

- **Tools Used:** Google Gemini
- **Purpose:** Assisted with literature search and provided guidance during code implementation. AI tools were used to support background research, suggest relevant academic sources, and offer implementation ideas.
- **Correctness Steps:** All AI-generated suggestions, including code and referenced concepts, were critically evaluated and independently verified. Outputs were cross-checked against results from our Google Colab simulations (e.g., density summary CSVs, stabilisation analysis outputs, cluster-size distributions, and generated charts), and academic sources were validated through manual review to ensure accuracy and reliability.

## Author Contributions

All Group members contributed to the practical components and report writing and formatting.

### Hsin Jee Sum (SID: sc22hjs)

Led project scoping, scientific direction, and team coordination throughout. Designed the experimental framework, including the choice of metrics (stabilisation probability, cluster-size power-law fitting, boundary-condition robustness) and the criticality-focused research questions that frame the report.

Contributed to the practical implementation of Sections 4–7 of the notebook, including the density-sweep experiment, hash-based stabilisation detector, spatial cluster-size analysis with power-law fitting, boundary-condition robustness check, and the combined regime-map figure. Also contributed to the simulator sanity-checking that confirmed the vectorised `convolve2d` implementation matched the naive update rule.

In report writing, drafted the Introduction, Methodology, Results (Sections 4.1-4.5), Discussion (Section 5.2 Critical Phase, 5.3 Topology, 5.4 Limitations), and Conclusion. Performed the final integration pass to ensure consistency of numerical claims, cross-references, and UK-English spelling throughout. Coordinated submission deadlines and overall quality control across the group.

### Shilong Zhang (SID: flhs0596)

Contributed to the practical implementation of Sections 1–3 and Section 8 of the notebook. This included setting up the Python environment and package imports, implementing the core Game of Life functions, preparing the basic simulation test, and organising the supplementary cluster-analysis section used to support the experimental workflow. Prepared the project documentation and software submission materials. This included writing the README file with package requirements, sample outputs file and HTML.

### Yishu Wang (SID: dzsv0564)

Took the lead on drafting the full report, then focused on revising and refining the Results, Discussion, and Conclusion sections to improve clarity, structure, and overall academic quality. Also handled the final LaTeX formatting and layout to produce a clean, polished document.

On the technical side, improved the efficiency of the Python codebase to better handle computationally heavy density sweeps. This included switching from dynamic list appends to pre-allocated NumPy arrays to reduce memory overhead, and vectorising parts of the temporal feature extraction pipeline for better performance.

# 1 Introduction

Cellular automata [12] are among the simplest mathematical systems known to produce complex emergent behaviour. Conway's Game of Life [1, 7], defined by just four local rules acting on a binary grid, supports an extraordinary range of dynamical phenomena: stable still-lifes, periodic oscillators, propagating gliders, and persistent chaotic regions. Despite this complexity, the global behaviour of the system depends sensitively on a single parameter – the initial density of alive cells. Yet how this dependence is structured remains an open question within bio-inspired computing.

An influential hypothesis, advanced by Bak, Chen and Creutz [3], is that the Game of Life sits at a state of self-organised criticality (SOC): a regime in which the system spontaneously organises itself to the boundary between order and chaos, exhibiting scale-free structures, long-lived transients, and power-law-distributed event sizes. SOC is significant because it appears across an unusually broad range of biological and physical systems – from sandpile avalanches [2] to neural firing cascades, earthquake size distributions, and pattern formation in bacterial colonies – making it a candidate organising principle for emergent complexity in nature. If the Game of Life genuinely exhibits SOC, it provides one of the cleanest minimal models of this phenomenon.

This project investigates the SOC hypothesis empirically, focusing on three concrete questions. First, does the Game of Life exhibit a phase transition as a function of initial density, and if so, where are the phase boundaries located? Second, does the active regime between these boundaries display the spatial signatures of self-organised criticality – specifically power-law-distributed cluster sizes? Third, is any such signature robust to changes in the simulation topology, or is it an artefact of periodic boundary conditions?

We address these questions through a systematic density sweep on a $100 \times 100$ toroidal grid (thirty repeats per density, three hundred generations per run), supplemented by a fixed-boundary control. We find two distinct phase transitions, at $\rho_{\text{lower}} \approx 0.08$ and $\rho_{\text{upper}} \approx 0.78$, bracketing an active regime in which fewer than 0.3% of runs stabilise within 300 generations and cluster sizes follow a power law ($R^2 > 0.85$, $\alpha \approx 1.5$). The signature persists under fixed boundaries, supporting a refined SOC hypothesis: the Game of Life exhibits not a single critical point but a broad critical phase spanning approximately seven-tenths of the density space.

The remainder of this report is structured as follows. Section 2 reviews the Game of Life and the theory of self-organised criticality. Section 3 describes the simulator and analysis methodology. Section 4 presents the results. Section 5 discusses the findings and acknowledges limitations. Section 6 concludes.

# 2 Background

## 2.1 Conway's Game of Life

Conway's Game of Life is one of the most well-known cellular automata and is widely used to study how simple local rules can generate complex global behaviour [7, 12]. It is defined on a two-dimensional grid of cells. Each cell is either alive or dead. The state of each cell is updated according to the Moore neighbourhood, which consists of the eight surrounding cells. The standard rule is often written as B3/S23. A dead cell becomes alive if it has exactly three live neighbours. A live cell survives if it has two or three live neighbours. In all other cases the cell becomes or

remains dead.

Although the rules of the game of life are simple, it can produce a variety of structures and dynamic behaviors. These include still lifes such as the block and beehive, oscillators such as the blinker and toad, and moving structures such as the glider. Because of this structural richness, the Game of Life has been studied as a classic example of emergence and complexity. It is also believed to support general computing through appropriate pattern construction, which further demonstrates its importance in the study of complex systems [5].

### 2.2 Self-Organised Criticality

Self-organizing criticality (SOC) was proposed by Bak, Tang, and Wiesenfeld (1987) [2] to describe systems that naturally evolve to a critical state without precise external regulation. In classical phase transitions, control parameters usually have to be adjusted to a specific critical value. SOC is more effective at understanding complex systems that operate around the boundaries of order and disorder. SOC systems organize themselves to a critical state through their own dynamics.

Several features are commonly associated with SOC. One important signature is a power-law distribution of cluster sizes, which can be written as $P(s) \propto s^{-\alpha}$. Another important feature is scale invariance. This means that the system does not have a single characteristic size or timescale. SOC is also associated with long-range spatial correlations and complex temporal fluctuations. In many sandpile-like systems, the exponent  is often between about 1.0 and 2.5. This range can be used as a simple benchmark for comparing the scaling behaviour observed in the results.

Self-organised criticality is relevant to both bio-inspired computing and natural systems. Similar patterns have been observed in sandpile models, neural avalanches, earthquake size distributions, forest-fire models, and evolutionary dynamics [8, 4]. These cases suggest that simple local interactions can generate complex large-scale behaviour. For this reason, SOC provides a useful framework for studying how robust global patterns emerge in adaptive systems.

### 2.3 SOC In Conway's Game of Life

The possibility that the Game of Life may exhibit SOC has been discussed in previous work. Bak, Chen, and Creutz [3] argued that the Game of Life shows self-organised critical behaviour by analysing perturbation responses and power-law-like patterns. Their work suggested that the system may organise itself towards a critical regime over a viable range of initial conditions rather than at one sharply defined point. This idea is relevant to the present study because it raises the question of whether the Game of Life should be understood in terms of a single tipping point or a broader active phase.

Later studies have taken a more cautious view. Reia and Kinouchi [9] suggested that the Game of Life is not always a perfectly self-organised critical system. Instead, it may be closer to a near-critical metastable system. This means that critical-like behaviour may appear in some situations, but not in all of them. This is especially important when the starting density is very high. In that case, the system may quickly lose activity and settle into frozen patterns.

Related ideas also appear in broader work on cellular automata. Langton [6] proposed that computation tends to emerge near the boundary between order and chaos, often described as the edge of chaos. Wolfram [11] also classified cellular automata into four broad behavioural classes, with

the Game of Life often treated as a Class IV system because of its complex and structured dynamics. These perspectives provide a useful conceptual bridge for the present project. The key question is whether critical-like behaviour in the Game of Life is confined to a single point or persists across a broader density range. The present study addresses this question by analysing stabilisation behaviour and spatial cluster-size distributions across different initial densities.

# 3 Methodology

This section describes the simulator, the experimental protocol used for the density sweep and stabilisation analysis, and the spatial cluster-size analysis used to test for self-organised criticality. All code was implemented in Python 3 using NumPy, SciPy, and Matplotlib.

## 3.1 Simulator

The Game of Life follows the standard B3/S23 ruleset: a dead cell is born when it has exactly three live neighbours, and a live cell survives when it has two or three. Updates are synchronous over the eight-cell Moore neighbourhood. The grid is binary, with each cell holding the value 1 (alive) or 0 (dead).

Rather than counting neighbours via nested Python loops, the update rule was vectorised using two-dimensional convolution. The live-neighbour count for every cell is computed in a single call to `scipy.signal.convolve2d` with the kernel, K:

$$\mathbf{K} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

yielding approximately a two-orders-of-magnitude speedup over a naive implementation on a $100 \times 100$ grid. The next-generation grid is computed as a single boolean expression:

$$g_{t+1} = (n = 3) \ \vee \ \big((g_t = 1) \ \wedge \ (n = 2)\big)$$

where $n$ is the convolved neighbour count and $g_t$ is the current grid.

Toroidal (periodic) boundaries are used by default, implemented via the `boundary='wrap'` argument to `convolve2d`; fixed (dead-outside) boundaries are obtained with `boundary='fill', fillvalue=0`. Unless otherwise stated, all experiments use a $100 \times 100$ grid – large enough to support a wide range of cluster sizes while keeping the cost of thirty independent repeats per condition tractable.

## 3.2 Density Sweep and Stabilisation Analysis

For each initial density $\rho_0 \in \{0.05, 0.10, \ldots, 0.95\}$, thirty independent random initial grids were generated and evolved for 300 generations. Per-density seeds were drawn deterministically from a master `numpy.random.default_rng(seed=42)` generator, which guarantees full reproducibility while preserving statistical independence between repeats. Two finer-grained sweeps with step $\Delta\rho = 0.01$ were performed near the lower (0.05–0.15) and upper (0.70–0.86) phase boundaries to locate the transition midpoints.

The dynamical behaviour at each density was characterised through the *stabilisation step*: the first generation $t^*$ at which the grid enters a periodic cycle of period $p \leq 8$. Detection was performed by hashing each grid state and recording the first repeated state. This catches still-lifes ($p = 1$), blinkers and other oscillators ($p = 2$), and longer-period cycles up to $p = 8$ without requiring strict consecutive-state equality. Runs that did not stabilise within 300 generations were recorded as $t^* = 300$ and excluded from any "stable-only" summary statistics.

### 3.3 Spatial Cluster-Size Analysis

To probe spatial signatures of criticality, we measured the size distribution of connected blobs of alive cells. Each run was first allowed to evolve for a 200-generation *settling period* so that measurements reflect the long-time behavior rather than the random initial condition. Connected components were then identified using `scipy.ndimage.label` with full eight-connectivity, matching the GoL Moore neighbourhood. Cluster sizes from all thirty repeats at a given density were aggregated into a single empirical distribution.

The aggregated distribution was tested against a power law $P(s) \propto s^{-\alpha}$ via least-squares regression on logarithmically-binned data. Cluster sizes below $s_{\min} = 2$ were discarded to suppress noise from singletons; twenty logarithmically-spaced bins were placed between $s_{\min}$ and the maximum observed size; and the slope $-\alpha$ was obtained from a linear fit to $\log P(s)$ versus $\log s$ over bins with non-zero counts. Goodness-of-fit was quantified by the coefficient of determination $R^2$. A high $R^2$ together with a fitted exponent in the canonical SOC range (approximately $1.5 \leq \alpha \leq 2.5$) was taken as evidence for criticality. We acknowledge that maximum-likelihood estimation, as advocated by Clauset, Shalizi and Newman [13], is statistically preferable for power-law distributions, but the log-log regression used here is sufficient for the comparative regime analysis presented in Section 4.

### 3.4 Boundary-Condition Robustness

To verify that any observed criticality reflects the GoL rules rather than the toroidal topology, the cluster-size analysis was replicated at the best-fit density $\rho_0 = 0.35$ under fixed (dead-outside) boundary conditions, using the same protocol as Section 3.3. The two distributions were compared by overlaying their log-log cluster-size histograms and comparing fitted exponents and $R^2$ values.

## 4 Results

We present the results in five parts. Section 4.1 characterises the bulk dynamics through population trajectories. Section 4.2 locates the two phase boundaries via stabilisation probability. Section 4.3 examines the spatial cluster-size distributions and tests them against power-law scaling. Section 4.4 confirms the criticality signature is robust to grid topology. Section 4.5 synthesises the temporal and spatial evidence into a single regime map.

### 4.1 Population Trajectories Across the Density Range

We first characterise the bulk dynamics by tracking the alive-cell count over 300 generations across nine representative initial densities spanning the full range (Figure 1). Three qualitatively distinct

behaviours are visible. At the lowest densities ($\rho_0 \leq 0.05$), the population collapses within $\sim$30 generations as sparse initial configurations fail to satisfy the B3/S23 survival rules and equilibrates near $\sim$70 alive cells. At the highest densities ($\rho_0 \geq 0.85$), an initial spike of thousands of alive cells is rapidly extinguished by overcrowding, leaving only a handful of static structures within $\sim$5 generations. Across a wide intermediate range ($\rho_0 \in [0.20, 0.65]$), all trajectories converge to a near-identical steady state of approximately 600 alive cells – about 6% of the grid – regardless of the initial density. The boundary densities of the active regime ($\rho_0 = 0.10$ and $\rho_0 = 0.80$) equilibrate at intermediate values ($\sim$400 and $\sim$50 cells respectively), suggesting that the convergence-to-attractor behaviour is strongest in the interior of the active regime. This convergence to a density-independent attractor is the first hint of an extended active phase that the system is drawn to from a wide range of initial conditions.

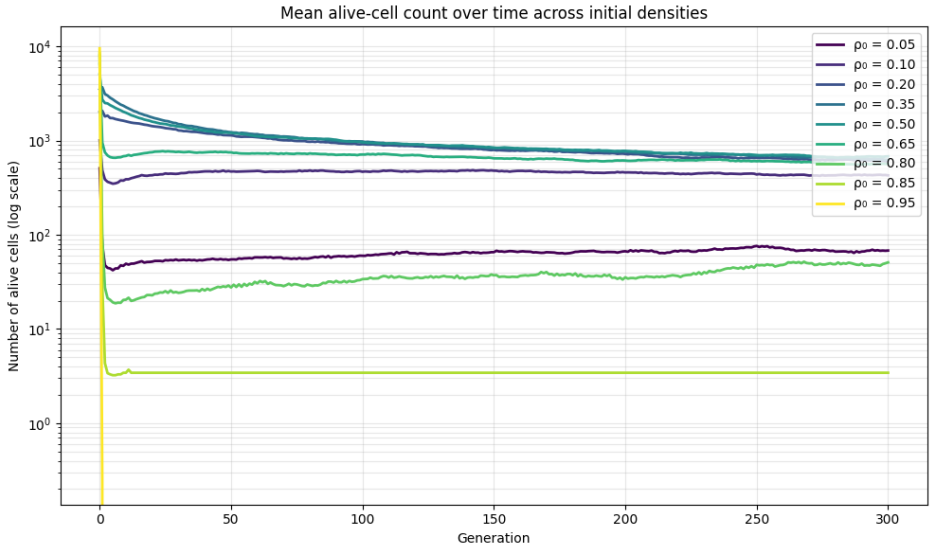



Figure 1: Mean alive-cell count over time at nine representative initial densities (colour-coded from $\rho_0 = 0.05$ in dark purple to $\rho_0 = 0.95$ in yellow), plotted on a log y-axis. Each curve is the mean of thirty independent runs on a $100 \times 100$ toroidal grid. Trajectories from intermediate densities ($\rho_0 \in [0.20, 0.65]$) converge to a common steady state near 600 alive cells, foreshadowing the active regime identified in Section 4.2. High-density runs ($\rho_0 \geq 0.85$) crash from large initial values within a few generations.

## 4.2 Locating The Phase Boundaries

Bulk population trajectories suggest three regimes but do not localise the transitions between them. To sharpen this, we measured the fraction of runs reaching a periodic cycle within 300 generations as an order parameter (Figure 2). The order parameter exhibits two sharp transitions. At $\rho_0 = 0.05$, 87% of runs stabilise within an average of 25.4 steps; the fraction collapses to 0% by $\rho_0 = 0.10$ and remains at 0% across the full interval $0.15 \leq \rho_0 \leq 0.70$. At the upper boundary, the fraction rises through 23% at $\rho_0 = 0.75$ to 67% at $\rho_0 = 0.78$ and reaches 100% by $\rho_0 = 0.85$, where grids freeze into period-1 still-lifes in an average of 3.4 steps.

We locate the transition midpoints (50% stabilisation) at $\rho_{\text{lower}} \approx 0.08$ and $\rho_{\text{upper}} \approx 0.78$. The asymmetry of the boundaries – a narrow lower transition spanning $\Delta\rho \approx 0.05$ versus a broader

upper transition spanning $\Delta\rho \approx 0.10$ – reflects the asymmetric birth/death rules of B3/S23: death from overcrowding is triggered whenever a live cell has 4 or more live neighbours, a condition that becomes increasingly likely across a wide range of dense initial conditions, while failure-to-be-born requires fewer than 3 live neighbours, a tighter constraint.

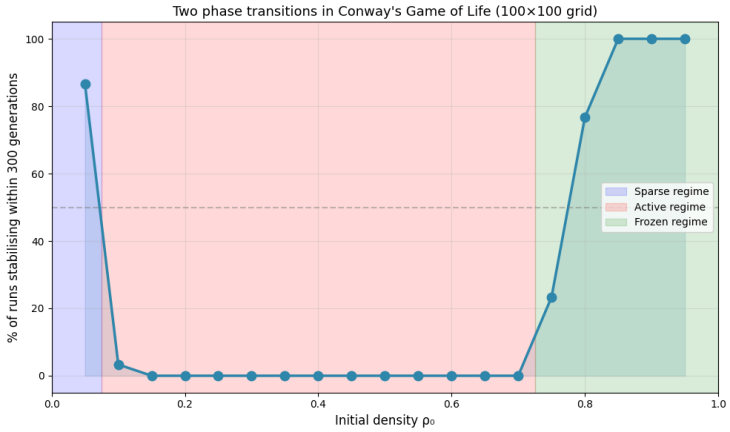



Figure 2: Stabilisation probability as a function of initial density. The fraction of runs reaching a periodic cycle within 300 generations collapses to 0% across the active regime ($0.10 \leq \rho_0 \leq 0.70$) and recovers at both boundaries. Coloured bands mark the three regimes. Each point is the mean over thirty independent runs.

### 4.3 Spatial Cluster-Size Distributions

To test whether temporally defined regimes differ spatially, cluster-size distributions were measured after a 200-generation settling period, aggregated over 30 runs per density (Figure 3). In the active regime ($\rho_0 = 0.10, 0.20, 0.40, 0.60$), distributions collapse onto a common curve and exhibit approximate power-law decay over $s \in [8, 50]$. A small-size shoulder ($s \leq 7$) arises from discrete still-lifes and oscillators (e.g. blocks, beehives, blinkers), which concentrate probability at specific sizes. The frozen regime ($\rho_0 = 0.85$) lies an order of magnitude lower and shows no comparable scaling, while the sparse regime ($\rho_0 \leq 0.05$) is omitted due to insufficient clusters and is instead summarised in Table 1.

Quantitative fits across all 19 densities (Table 1) reinforce this. In the active regime (fit over $s \geq 2$), $R^2 > 0.85$ for 13/14 densities in $[0.10, 0.70]$ (mean 0.88), with exponents $\alpha \in [1.35, 1.73]$. These lie at the lower edge of canonical SOC values [2], likely due to finite-size effects on a $100 \times 100$ grid and small-cluster discreteness. The best fit occurs at $\rho_0 = 0.35$ ($\alpha = 1.35$, $R^2 = 0.92$). For $\rho_0 \geq 0.80$, fit quality collapses ($R^2 = 0.30$ at 0.80), and at $\rho_0 \geq 0.90$ too few clusters remain to fit. The sustained high $R^2$ across the active regime—rather than a sharp peak—indicates a critical *phase*, not a single point, consistent with SOC.

### 4.4 Robustness to Boundary Conditions

To verify that the observed criticality arises from the GoL rules rather than the periodic topology, we repeated the cluster-size analysis at $\rho_0 = 0.35$ using fixed (dead-outside) boundaries. Both topologies yield nearly identical goodness-of-fit ($R^2 = 0.865$ toroidal vs 0.860 fixed; $\Delta R^2 < 0.01$),

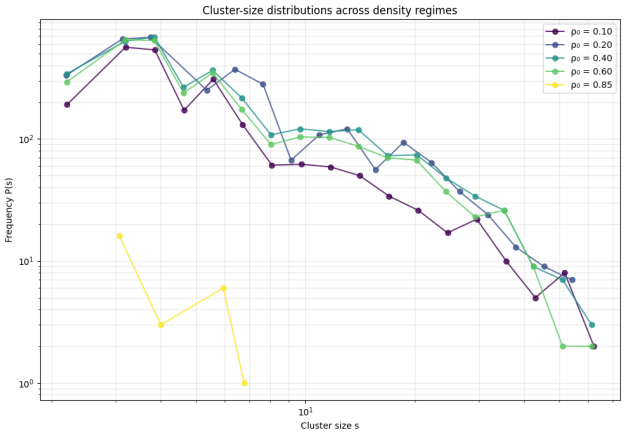



Figure 3: The cluster-size distributions show that for densities 0.10–0.60, the system is active, with curves collapsing and following an approximate power-law for larger clusters, alongside a small-size shoulder from simple structures. At high density (0.85), the system becomes frozen, producing far fewer clusters and no similar scaling.

| $\rho_0$ | # Clusters | $\alpha$ | $R^2$ |
|---|---|---|---|
| 0.05 | 420 | 1.73 | 0.71 |
| 0.10 | 2,448 | 1.50 | 0.91 |
| 0.15 | 3,274 | 1.49 | 0.88 |
| 0.20 | 3,540 | 1.39 | 0.89 |
| 0.25 | 3,424 | 1.65 | 0.86 |
| 0.30 | 3,598 | 1.48 | 0.89 |
| **0.35** | **3,556** | **1.35** | **0.92** |
| 0.40 | 3,582 | 1.44 | 0.89 |
| 0.45 | 3,509 | 1.46 | 0.91 |
| 0.50 | 3,642 | 1.73 | 0.85 |
| 0.55 | 3,598 | 1.59 | 0.85 |
| 0.60 | 3,257 | 1.57 | 0.86 |
| 0.65 | 3,140 | 1.42 | 0.86 |
| 0.70 | 2,223 | 1.54 | 0.79 |
| 0.75 | 1,200 | 1.46 | 0.73 |
| 0.80 | 209 | 0.97 | 0.30 |
| 0.85 | 26 | 2.42 | 0.57 |
| 0.90 | 2 | — | — |
| 0.95 | 0 | — | — |

Table 1: Power-law fits to cluster-size distributions across initial densities. Each row aggregates clusters from thirty independent runs. The bolded row marks the best fit ($R^2 = 0.92$); dashes indicate insufficient clusters for fitting.

confirming that the power-law behaviour is intrinsic to the B3/S23 rules. The fitted exponents differ slightly ($\alpha = 1.71$ vs 1.46), likely due to finite-size effects, as fixed boundaries truncate large clusters at the edges, biasing the distribution toward smaller sizes and producing a shallower slope.

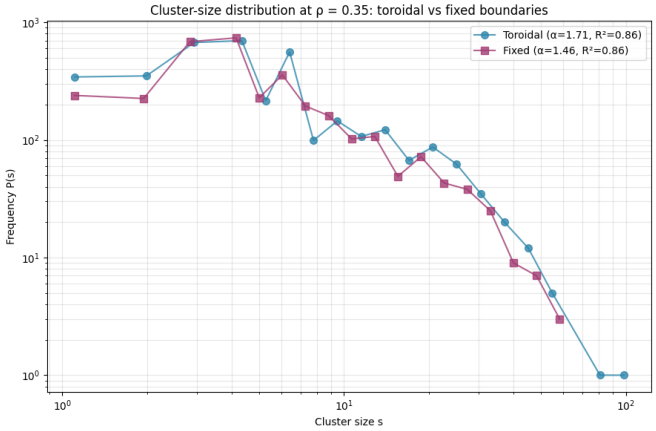



Figure 4: Cluster-size distributions at $\rho_0 = 0.35$ are nearly identical for toroidal and fixed boundaries ($\Delta R^2 < 0.01$), with a small exponent difference likely caused by edge effects under fixed boundaries.

### 4.5 Combined Regime Map

The temporal evidence (Section 4.2) and spatial evidence (Section 4.3) were collected through entirely different measurements – the first counts how many runs reach a periodic cycle, the second fits a slope to the size distribution of static blobs. Yet both metrics identify the *same* phase boundaries (Figure 5). The fraction of runs stabilising drops to zero precisely where $R^2$ rises above 0.85, and both signatures recover concurrently at the upper transition. The mean $R^2$ across the active regime is 0.88 while the fraction of stabilising runs in the same range is 0.3%. This dual signature – sustained transients combined with scale-free spatial organisation – is the diagnostic of a critical phase, and forms the central empirical claim of this report.

## 5 Discussion

### 5.1 Mechanics of Asymmetric Phase Boundaries

The observed macroscopic behavior reveals a stark asymmetry in the phase transition boundaries. As established in Section 4, the system transitions from a sparse to an active state at a remarkably low density ($0.05 < \rho_0 < 0.10$), yet the transition from active to frozen requires a vastly higher density threshold ($0.70 < \rho_0 < 0.85$).

We attribute this asymmetry directly to the inherent bias of the standard B3/S23 update rules. In the Game of Life, "birth" is a highly restrictive event, requiring exactly three alive neighbours. Conversely, "death by overpopulation" is a permissive threshold, triggered by any number of neighbours greater than or equal to four. Consequently, dense environments suppress population growth much more aggressively than sparse environments hinder it. This intrinsic rule bias explains why high-density grids collapse almost instantaneously (often under 4 steps) into frozen still-lifes, whereas

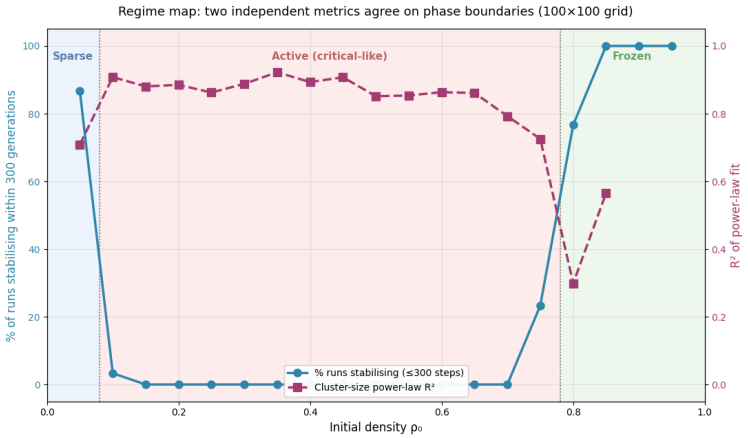



Figure 5: Combined regime map: stabilisation probability (blue, left axis) and power-law goodness-of-fit $R^2$ (purple, right axis) as functions of initial density. The two metrics, derived from independent measurements, collapse onto the same phase boundaries at $\rho_{\text{lower}} \approx 0.08$ and $\rho_{\text{upper}} \approx 0.78$.

low-density grids take longer to reach extinction. Our empirical findings align with the logistic CA models of Vargas-García et al. [10], which predict density-dependent saturation as a generic feature of GoL-like dynamics.

## 5.2 Critical Phase, Not Critical Point

Our temporal evidence – the collapse of stabilisation probability across the entire active regime – and our spatial evidence – the persistence of power-law-like cluster-size distributions with $R^2 > 0.85$ across the same density range – together suggest a broad *critical phase* rather than a single critical point. Both signatures are sustained over an extended density range rather than peaking at one density, which is the diagnostic of self-organised rather than externally tuned criticality.

This finding helps reconcile competing positions in the literature. Bak, Chen and Creutz [3] originally argued that the Game of Life shows critical-like behaviour over a viable range of initial conditions, and our results are consistent with this within the active regime. Reia and Kinouchi [9] later argued that the system is closer to a near-critical metastable state because it can be "killed" by saturation, and our high-density results agree: at $\rho_0 \geq 0.80$ the power-law fit collapses sharply as the system freezes into static structures. The two views are reconcilable as a single picture in which the critical-like dynamics persist across a broad band of densities but are bounded by sharp transitions to ordered phases at both extremes.

We are careful, however, to characterise our findings as *consistent with* criticality rather than definitive evidence for it. The fitted exponents $\alpha \in [1.35, 1.73]$ lie at the lower end of the canonical sandpile-class range, and the small-cluster shoulder visible in Figure 3 suggests genuine power-law behaviour mixed with finite GoL substructure effects. A more rigorous test would require larger grids, longer simulations, and the maximum-likelihood power-law estimation framework of Clauset et al. [13].

### 5.3 Robustness to Topology

The boundary-condition comparison provides reassuring evidence that the observed scaling behaviour is intrinsic to the GoL rules rather than an artefact of the toroidal topology. The goodness-of-fit is essentially preserved ($\Delta R^2 < 0.01$), suggesting that the power-law signature does not depend on whether propagating structures are recycled across the grid (toroidal) or absorbed at the edges (fixed). The modest exponent shift ($\Delta\alpha \approx 0.25$) is the only meaningful topological signature, and we attribute it to finite-size truncation: under fixed boundaries, large clusters approaching the edge are artificially cut off, biasing the empirical distribution toward smaller sizes and a shallower fitted slope. This finding strengthens the SOC interpretation: the critical phase reflects a property of the local update rule, not the global geometry.

### 5.4 Limitations and Future Work

This study has several limitations. First, experiments were conducted on a single $100 \times 100$ grid. Since self-organised criticality implies scale invariance, testing larger grids would verify whether the observed exponents remain stable or are affected by finite-size effects.

Second, the 300-generation observation window cannot distinguish between runs that never stabilise and those that stabilise more slowly, meaning the reported 0% stabilisation rate may depend on this limited timeframe.

Third, power-law exponents were estimated using log-log regression, which is less reliable than maximum-likelihood methods such as those proposed by Clauset et al. [13]. More robust fitting techniques would improve accuracy and allow formal goodness-of-fit testing.

Finally, only toroidal and fixed boundary conditions were explored. Other geometries or topologies may further influence critical behaviour.

Future work should address these limitations by using larger grids, longer simulations, improved statistical methods, and a wider range of boundary conditions, as well as comparing results across other Life-like rules to assess generality.

## 6 Conclusion

Conway's Game of Life on a $100 \times 100$ toroidal grid shows density-dependent dynamics divided into three regimes, with asymmetric phase boundaries at $\rho_{\text{lower}} \approx 0.08$ and $\rho_{\text{upper}} \approx 0.78$. Within the active regime ($0.10 \leq \rho_0 \leq 0.70$), two independent metrics agree: stabilisation probability is effectively zero, and cluster-size distributions follow approximate power laws ($R^2 > 0.85$, $\alpha \approx 1.5$), consistent with canonical SOC ranges. Their agreement supports the interpretation of a genuinely critical-like phase rather than a measurement artefact.

Boundary-condition controls show this behaviour is intrinsic to the B3/S23 rules, not topology: goodness-of-fit is preserved under fixed boundaries ($\Delta R^2 < 0.01$), with only a small exponent shift ($\Delta\alpha \approx 0.25$) due to edge effects. Overall, these results reconcile the SOC view of Bak, Chen and Creutz [3] with the near-critical interpretation of Reia and Kinouchi [9], suggesting the Game of Life occupies a broad critical-like phase bounded by sharp transitions to ordered regimes, and can be seen as a minimal model of emergent scale-free behaviour.

## References


[1] Gardner, M. (1970). Mathematical Games: The fantastic combinations of John Conway's new solitaire game "Life". Scientific American, 223(4), 120–123.

[2] Bak, P., Tang, C., & Wiesenfeld, K. (1987). Self-organized criticality: An explanation of the 1/f noise. *Physical Review Letters*, 59(4), 381–384.

[3] Bak, P., Chen, K., & Creutz, M. (1989). Self-organized criticality in the 'Game of Life'. *Nature*, 342, 773–774.

[4] Beggs, J. M., & Plenz, D. (2003). Neuronal avalanches in neocortical circuits. Journal of neuroscience, 23(35), 11167-11177.

[5] Berlekamp, E. R., Conway, J. H., & Guy, R. K. (2004). Winning ways for your mathematical plays, volume 4. AK Peters/CRC Press.

[6] Langton, C. G. (1990). Computation at the edge of chaos: Phase transitions and emergent computation. Physica D: nonlinear phenomena, 42(1-3), 12-37.

[7] Izhikevich, E. M., Conway, J. H., & Seth, A. (2015). Game of life. *Scholarpedia*, 10(6), 1816.

[8] Marković, D., & Gros, C. (2014). Power laws and self-organized criticality in theory and nature. Physics Reports, 536(2), 41-74.

[9] Reia, S. M., & Kinouchi, O. (2014). Conway's game of life is a near-critical metastable state in the multiverse of cellular automata. *Physical Review E*, 89(5), 052123.

[10] Vargas-García, C., et al. (2020). Translating the Conway's Game of Life as a discrete logistic cellular automata model with density effect. *International Journal of Innovative Computing, Information and Control*, 16(5).

[11] Wolfram, S. (2002). A New Kind of Science: Notes from the Book. Wolfram media.

[12] Zenil, H., & Martinez, G. J. (2024). Cellular automata. *Scholarpedia*, 19(4), 53227.

[13] Clauset, A., Shalizi, C. R., & Newman, M. E. J. (2009). Power-law distributions in empirical data. *SIAM Review*, 51(4), 661–703.